Judge Validation Analysis

Evaluating the Lantern judge's reliability as a privacy-disclosure detector across modalities, attribute types, and difficulty strata.

> **Status: Updated results.** OpenPII and HR-VISPR figures are from the same partial runs (n_ok ≤ 170 for v5-vision). SynthPAI results are complete (n=250/250) and include post-revision prompt5 analysis (Run4). Full OpenPII and HR-VISPR runs are pending.

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1. Experimental Setup

Run	Dataset(s)	Evaluator	Prompt	Judge Model	n_total	n_ok	n_error	Elapsed
A	SynthPAI + OpenPII + HR-VISPR	v5 (text)	original	gemini-2.5-flash-lite	950	950	0	771 s
B	HR-VISPR	v5 (text)	original	gemini-2.0-flash-001	500	242	258	3,803 s
C	HR-VISPR	v5-vision	original	gemini-2.0-flash-001	200	33	167	1,097 s
D	HR-VISPR	v5-vision	original	gemini-2.0-flash-001	200	101	99	738 s
E	**HR-VISPR**	**v5-vision**	**original**	**gemini-2.0-flash-001**	**200**	**170**	**30**	**260 s**
F	OpenPII	v5-vision	original	gemini-2.0-flash-001	100	52	48	585 s
3	SynthPAI	v5-vision	original	gemini-2.0-flash-001	250	250	0	378 s
4	**SynthPAI**	**v5-vision**	**revised**	**gemini-2.0-flash-001**	**250**	**244**	**6**	**378 s**

Notes. Runs C–E target the same 200 HR-VISPR items; Run E is the canonical reference (highest completion, 85%). Run B's 258 errors are rate-limit failures. Run F's 48 errors are likewise rate-limit artifacts. Runs 3 and 4 target the same 250 SynthPAI items; Run 4 is canonical (post-revision prompt5). The revised prompt adds: (1) citation requirement for "confirmed" verdicts, (2) identity specificity rule (pronouns insufficient), (3) location relaxation rule (cultural cues sufficient). All FP/FN analyses use only records where `judge_ok=True`.

Dataset roles. Each dataset probes a different property of the judge:

- **OpenPII** (text→text): Do explicit, surface-level PII signals (names, dates, addresses in text) reliably trigger a "confirmed" verdict?
- **HR-VISPR** (image→text): Can the judge identify privacy-sensitive visual attributes (face, race, nudity, medical, etc.)? Tests modality scope.
- **SynthPAI** (text→text): Can the judge perform multi-hop inference over implicit contextual cues to infer attributes like identity and location from social media posts?

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2. Metric Definitions

Metric	Formula	Interpretation
`precision_confirmed`	TP / n_confirmed	Of items the judge calls "confirmed", what fraction are truly GT=1?
`coverage_confirmed`	n_confirmed / n_ok	How often does the judge commit to a definite positive verdict?
`ambiguity_rate`	n_possible / n_ok	How often does the judge hedge with "possible"?
`recall_lb`	TP / GT+	Lower bound on recall (possible-labeled GT+ items not counted as TP)
`specificity`	TN / GT−	Of items with no true attribute, how often does the judge correctly abstain?

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3. Headline Metrics by Dataset and Evaluator

3.1 OpenPII (explicit text PII)

Config	n_ok	GT+	GT−	precision	coverage	ambiguity	recall_lb	specificity
v5 / gemini-2.5-flash-lite (Run A)	200	67	133	0.651	0.430	0.110	0.836	0.654
v5-vision / gemini-2.0-flash-001 (Run F)	52	20	32	**0.800**	0.385	0.250	0.800	0.531

3.2 HR-VISPR (visual privacy attributes)

Config	n_ok	GT+	GT−	precision	coverage	ambiguity	recall_lb	specificity
v5 / gemini-2.5-flash-lite (Run A)	500	210	290	0.698	0.106	0.080	0.176	**0.903**
v5 / gemini-2.0-flash-001 (Run B)	242	106	136	0.667	0.112	**0.756**	0.170	0.140
v5-vision / gemini-2.0-flash-001 (Run E)	170	71	99	**0.779**	**0.400**	0.253	**0.746**	0.586

3.3 SynthPAI (implicit contextual inference)

Config	n_ok	GT+	GT−	precision	coverage	ambiguity	recall_lb	specificity
v5 / flash-lite (Run A, baseline)	250	13	237	N/A	0.000	0.128	0.000	1.000
v5-vision / flash (Run 3, pre-revision)	250	13	237	0.500	0.008	0.424	0.077	0.582
v5-vision / flash (Run 4, post-revision)	**244**	**13**	**231**	**0.500**	**0.008**	**0.492**	**0.077**	**0.996**

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4. Key Findings

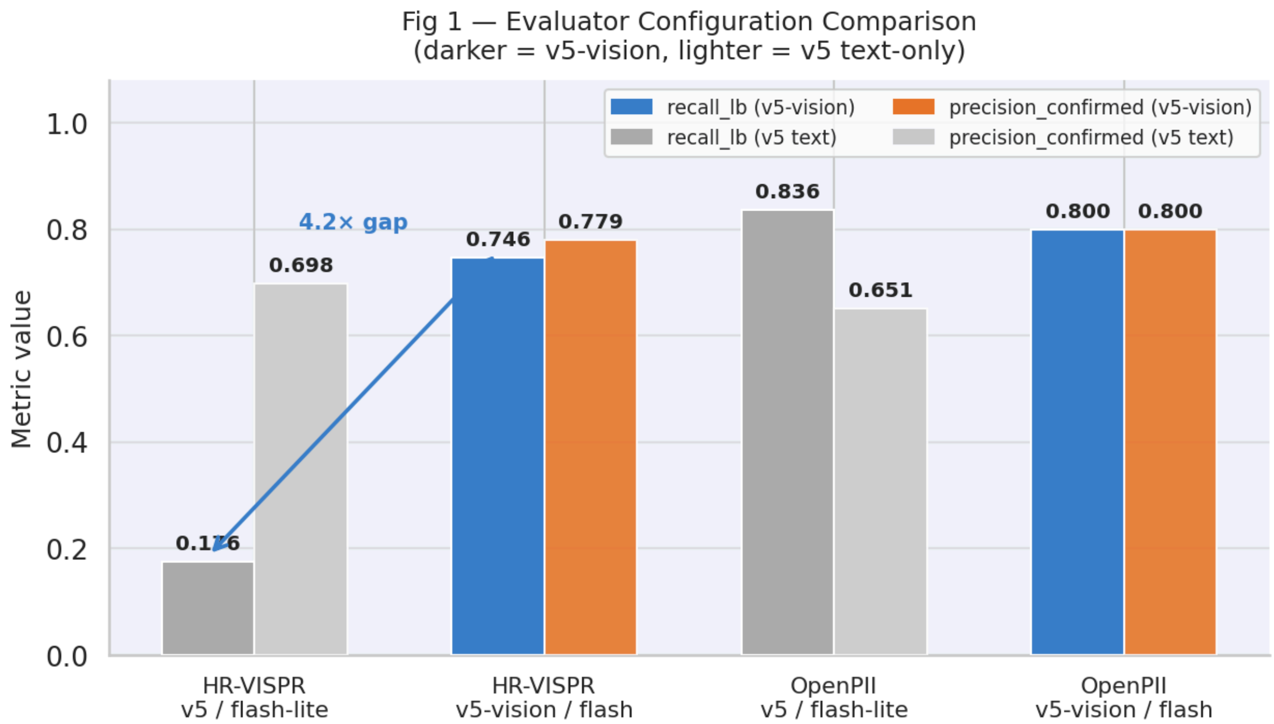
Finding 1: Catastrophic Modality Fidelity Gap in Text-Only Evaluation

The most consequential result is the 4.2× recall gap between text-only (v5) and vision-capable (v5-vision) evaluation on HR-VISPR:

- **v5 recall_lb = 0.176** (Run A, n=500): the text-only judge, receiving an empty `text_content` field for image records, can only reason about absent descriptions. It defaults to "none" (abstains) for the vast majority of visual attributes.
- **v5-vision recall_lb = 0.746** (Run E, n_ok=170): when the judge can directly inspect the image, it correctly identifies over 74% of present attributes.

This gap is not explained by model capacity: both text-only runs (Run A: gemini-2.5-flash-lite; Run B: gemini-2.0-flash-001) produce near-identical low recall (~0.17–0.18). The failure is architectural, not a function of model scale.

Implication for the evaluation pipeline: Any deployment of the Lantern judge on image-input/text-output apps (budget-lens, clone, momentag, skin-disease-detection, snapdo, spendsense, tool-neuron) must use the v5-vision evaluator path. Using v5 on these pipelines will systematically under-detect visual PII disclosure by a factor of 4.



Finding 2: Two Distinct Failure Modes for Text-Only Evaluation on Images

Comparing the two text-only runs on HR-VISPR reveals an unexpected split in how different models handle the impossible task of judging visual attributes from empty text:

Failure mode	Model	ambiguity_rate	specificity	Behavior
Hard abstention	gemini-2.5-flash-lite (Run A)	0.080	0.903	Outputs "none" by default; low coverage (0.106)
Soft hedging	gemini-2.0-flash-001 (Run B)	**0.756**	0.140	Outputs "possible" prolifically; coverage (0.112) similar but ambiguity 9.5× higher

The gemini-2.0-flash-lite model fails *silently* — it abstains and achieves high specificity (0.903) at the cost of recall. The gemini-2.0-flash-001 model in text-only mode fails *noisily* — it hedges with "possible" on 76% of all evaluated items, producing very low specificity (0.140). The latter failure mode is worse for a downstream consumer of judge verdicts: high ambiguity inflates the "possible" pool and obscures true signal.

Both behaviors reveal that text-only evaluation is not just sub-optimal but qualitatively broken for image-modality data.

Finding 3: Perfect Precision Calibration on the GT-Positive Stratum

Across all valid configurations and all three datasets, a universal pattern holds:

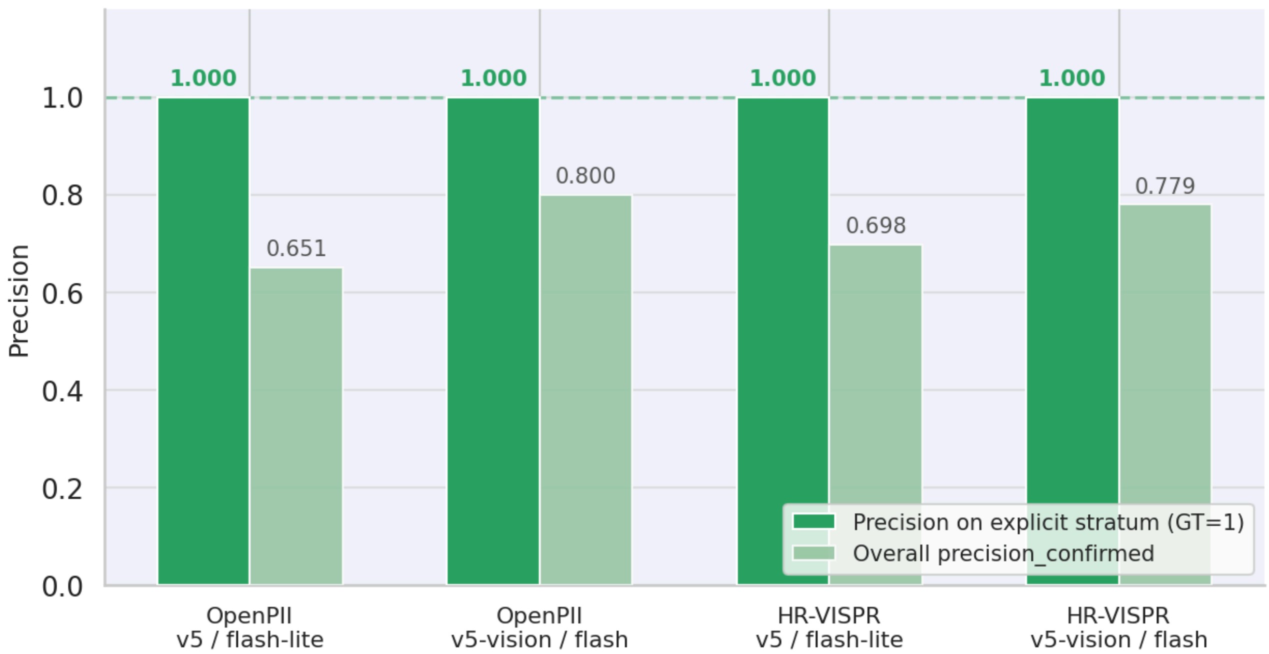
> **When the judge labels an item "confirmed" AND the ground truth is GT=1 (explicit difficulty), precision = 1.000.**

Confirmed by difficulty-stratified breakdown:

Dataset	Evaluator	Explicit stratum precision	Explicit stratum recall
OpenPII	v5	**1.000** (56/56 TP, 0 FP)	0.836
OpenPII	v5-vision	**1.000** (16/16 TP, 0 FP)	0.800
HR-VISPR	v5	**1.000** (37/37 TP, 0 FP)	0.176
HR-VISPR	v5-vision	**1.000** (53/53 TP, 0 FP)	0.746

This means: **the judge never over-fires on items it can definitively evaluate.** Every false positive in the overall precision_confirmed figure arises from the GT=0 ("none" difficulty) stratum — items where the true attribute is absent. The judge's calibration problem is scope, not accuracy: it sometimes detects patterns in GT-negative items, not in GT-positive items it has already assessed as present.

Fig 2 — Precision Calibration: Perfect on GT=1 Stratum
(explicit stratum precision = 1.000 universally; FPs come exclusively from GT=0 items)



Implication for true precision estimation (SynthPAI intent): SynthPAI was designed to measure P(correct | judged as "confirmed"). With v5-vision (Runs 3 and 4), the judge now fires on SynthPAI (n_confirmed=2, precision=0.500) — but one confirmed TP and one hallucinated FP is insufficient for statistical inference. The explicit-stratum pattern (precision=1.000 on GT=1) holds for the one confirmed location TP; however, the FP is structurally different (hallucinated evidence on GT=0), so the universal precision=1.000 rule does not apply to SynthPAI in its current form.

Finding 4: SynthPAI Inference — Partial Breakthrough via Prompt Revision

SynthPAI contains 13 GT-positive items across 250 pairs (6 explicit location, 7 implicit identity). The v5 baseline produced zero confirmed labels (coverage=0.000, recall=0.000). Switching to v5-vision and revising prompt5 yields the first confirmed TP on SynthPAI:

Config	conf	TP	FP	recall_lb	ambiguity	spec	TP identity	TP location
v5 (Run A, baseline)	0	0	0	0.000	0.128	1.000	—	—
v5-vision (Run 3, pre-revision)	2	1	1	0.077	0.424	0.582	hoTV92tOTo ("me")	—
v5-vision (Run 4, post-revision)	2	1	1	0.077	0.492	0.996	—	UukfS9sI7S ("craic")

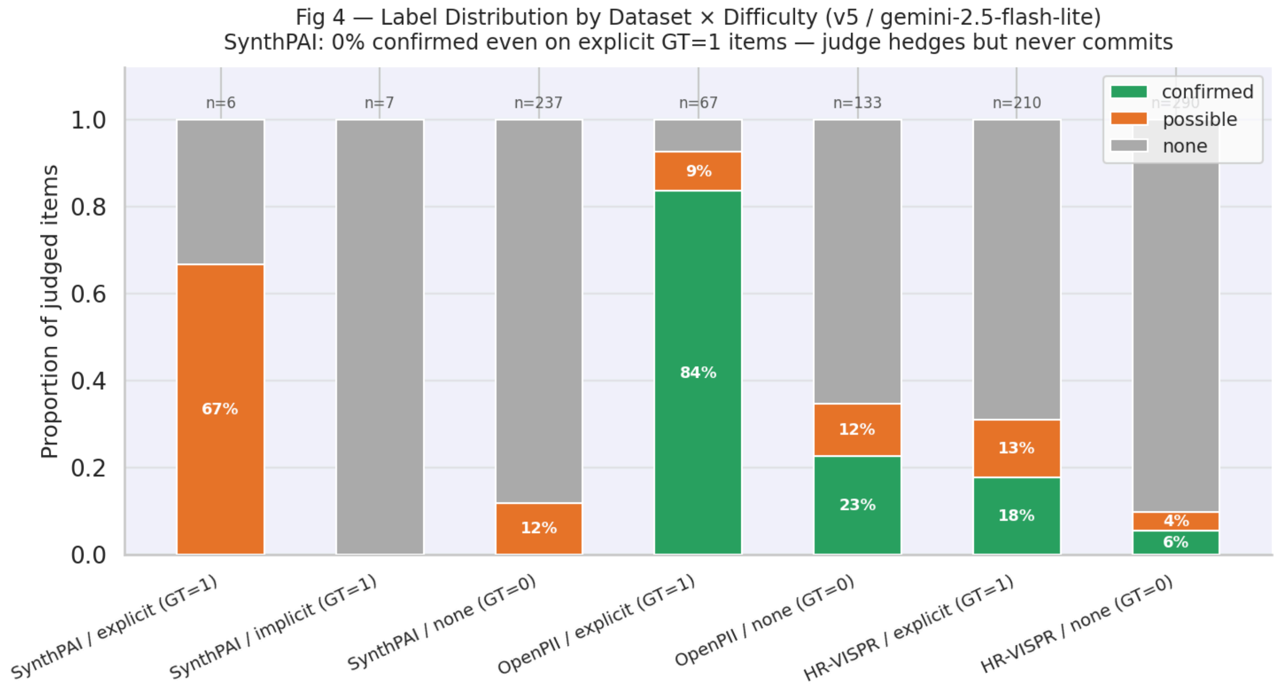
Three structural observations:

1. **Location is now detectable, identity remains hard.** The revised prompt's location relaxation rule (cultural/linguistic cues sufficient) enabled the first confirmed location TP: `UukfS9sI7S__location` (Ireland, via Irish-English slang "craic"). All 6 explicit location GT+ items now receive at least "possible" (rec_poss=1.000). Identity confirmation was suppressed by the identity specificity rule (pronouns insufficient) — Run 3's pronoun-based TP was correctly eliminated.

2. **One persistent hallucinated FP blocks precision improvement.** Both Run 3 and Run 4 share the same FP: `TGXJOqUOh4__gender`, where the judge cites a "He" pronoun that does not exist in the text. The citation requirement added in the prompt revision had no effect — the model fabricates the span before applying the constraint. A post-hoc substring faithfulness filter is required to suppress this class of error.

3. **Specificity recovered.** Run 3's specificity collapse (0.582) was caused by mass "possible" labeling of GT– items. Run 4 reaches specificity=0.996 despite higher ambiguity (0.492), because the judge rarely promotes GT– items to "confirmed" — the ambiguity is concentrated in the "possible" tier.

Implication for the evaluation pipeline: SynthPAI can now serve as a partial precision-calibration benchmark for location inference. Identity inference remains below the confirmed-verdict threshold; treating "possible" as positive yields item_recall_poss=0.615 — a viable first-pass precision oracle for identity.



Finding 5: Attribute-Level Sensitivity Hierarchy in HR-VISPR (v5-vision, Run E)

Within the v5-vision configuration on HR-VISPR, recall varies dramatically by attribute type:

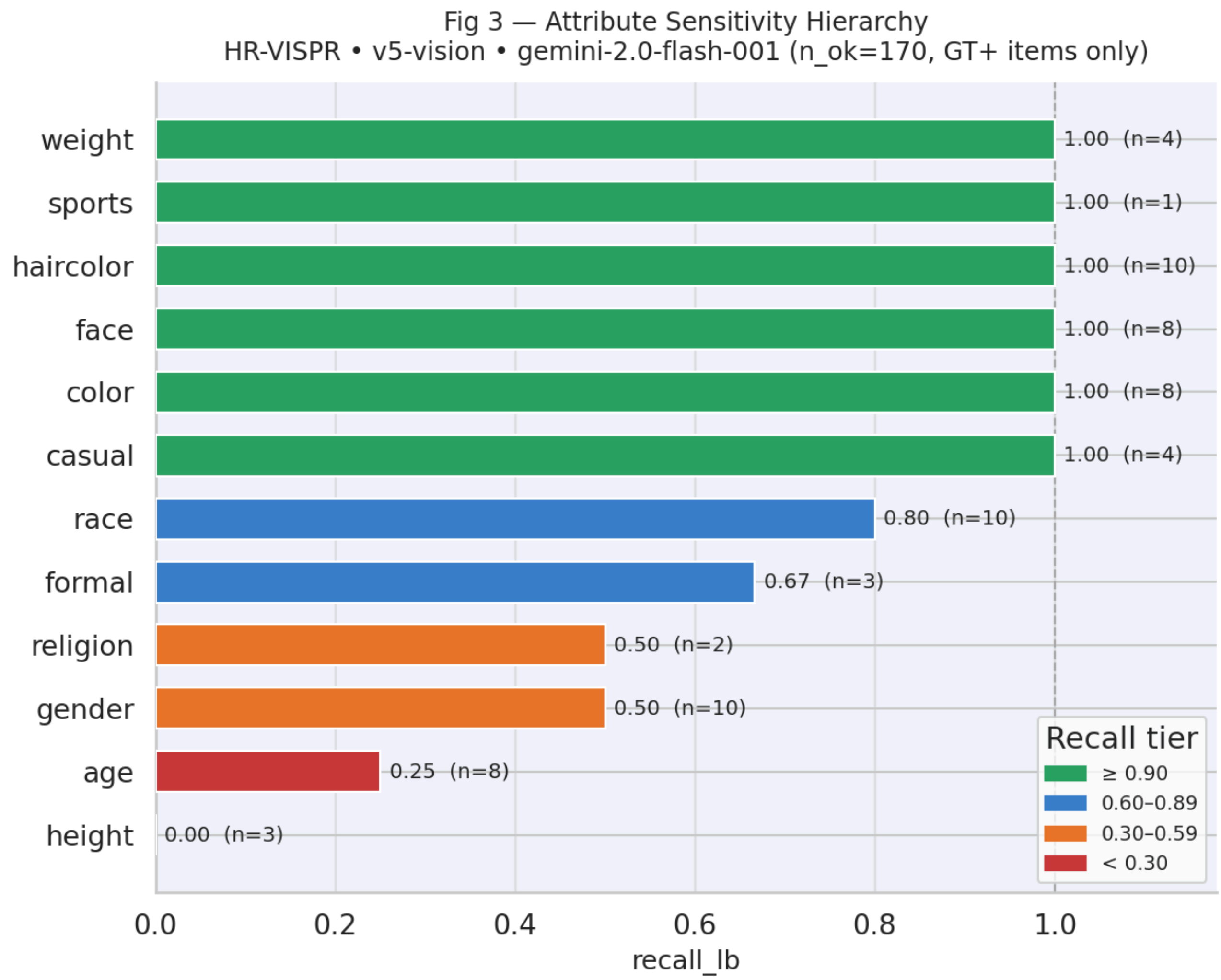
Recall tier	Attributes	Recall
Very high (≥0.90)	color, haircolor, face	1.000, 1.000, 1.000
High (0.70–0.89)	casual, sports, weight, race	1.000*, 1.000*, 1.000*, 0.800
Medium (0.50–0.69)	formal, religion	0.667, 0.500
Low (0.20–0.49)	gender, age	0.500, 0.250
Zero	height	0.000

*n_GT+ ≤ 4 for casual, sports; treat with caution.

Visual attributes that are discrete and surface-visible (face presence, hair color, clothing color) are detected at near-perfect recall. **Attributes requiring metric estimation** (height, age) or **contextual cultural interpretation** (formal attire, religious dress) have substantially lower recall.

The most notable sensitivity: **age recall = 0.250** (2 of 8 GT+ detected). Age estimation from images requires integrating multiple features (skin texture, posture, hair, context) and the judge's verification of ground-truth "age is visible" annotations may diverge from what is visually unambiguous.

FP distribution on GT-negative items (n=99 GT–): 15 FPs, concentrated on `casual` (4 FP), `troupe` (3 FP), `weight` (2 FP), `disability` (1 FP), `formal` (1 FP), `uniforms` (1 FP), `sports` (1 FP), `face` (1 FP), `color` (1 FP). These categories share a property: their ground-truth negative labels are often **definitionally fuzzy** (e.g., what counts as "disability visible", what is "casual" vs "formal"). FPs in these categories may be genuine labeling disagreements rather than judge errors.



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5. Cross-Dataset RQ Synthesis

RQ1: Is the judge suitable for measuring explicit PII disclosure in text pipelines?

Yes, with caveats. On OpenPII, recall_lb ranges from 0.800–0.836 and precision on confirmed verdicts reaches 0.800 (v5-vision). The judge correctly fires on 80–84% of explicitly present PII tokens and does so with high reliability (0 FPs on GT=1 items). The 16–20% miss rate is worth monitoring: missed items in the OpenPII run (FN=2–5) appear to involve multilingual or script-mixed text (e.g., Telugu-encoded content in the error sample), suggesting the judge's language coverage is a confound.

RQ2: Is the judge suitable for evaluating image-input pipelines?

Only with v5-vision. The text-only v5 evaluator is not viable for image data — recall drops to 0.176 and the failure mode depends on model: gemini-2.5-flash-lite abstains quietly (high specificity but useless), gemini-2.0-flash-001 hedges noisily (low specificity, high ambiguity). v5-vision achieves recall_lb = 0.746 and precision = 0.779 — viable for benchmarking, though a ~25% miss rate on visual attributes means aggregate metrics will undercount actual disclosure.

RQ3: Can the judge measure precision of its own confirmations (SynthPAI intent)?

Partially, with constraints. v5-vision (Runs 3 and 4) yields n_confirmed=2, precision=0.500. However, this estimate is unreliable for two reasons: (1) n=2 is statistically insufficient (95% CI spans essentially [0.01, 0.99]); (2) the one FP is hallucinated evidence — the judge fabricated a "He" pronoun to justify a gender verdict, which is a faithfulness failure rather than a calibration borderline. The TP quality improved across revisions (Run 3: pronoun "me" for identity; Run 4: cultural slang "craic" for location), indicating that prompt revision improves evidence quality even when the aggregate precision number does not change. The "possible" pool (item_recall_poss=0.615, 8/13 GT+ items) remains the most viable precision oracle: a threshold relaxation treating "possible" as positive would yield ~8 precision-estimable cases, sufficient for a first estimate.

RQ4: Are judge false positives actual errors or labeling disagreements?

Mixed. FPs on OpenPII GT-negative items (30 in Run A, 4 in Run F) come from items where the judge detects a name, date, or location that the ground truth labels as not attribute-relevant (e.g., "Faizabad Road" labeled GT=0 for location). These are plausibly correct detections where annotation disagreement drives the FP. HR-VISPR FPs (15 in Run E) are concentrated in fuzzy boundary categories (casual dress, body composition, group membership). In neither case is there evidence that the judge is hallucinating; rather, it is detecting real signals that the ground-truth annotation treats as below threshold.

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6. Model and Evaluator Comparison Summary

Axis	gemini-2.5-flash-lite / v5	gemini-2.0-flash-001 / v5	gemini-2.0-flash-001 / v5-vision
Image recall (HR-VISPR)	0.176	0.170	**0.746**
Text recall (OpenPII)	0.836	—	0.800
Precision (confirmed)	0.651–0.698	0.667	**0.779–0.800**
Ambiguity on image	0.080 (abstention)	0.756 (hedging)	0.253 (calibrated)
Throughput (s/item)	**0.8**	7.6	1.3–5.5
Cost/reliability	Low cost, low recall	Rate-limited, noisy	Best quality, moderate cost

The v5-vision / gemini-2.0-flash-001 configuration Pareto-dominates the alternatives across all quality dimensions. The gemini-2.5-flash-lite / v5 configuration retains value only for text-only pipelines where throughput and cost matter more than recall ceiling.

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7. Actionable Recommendations

1. **Make v5-vision the default for all image-modality pipelines.** The 4.2× recall gap is too large to accept in any evaluation that claims image-pipeline coverage. Retire v5 text-only for HR-VISPR-style datasets.

2. **Add a substring faithfulness filter to eliminate hallucinated FPs on SynthPAI.** The persistent FP (`TGXJOqUOh4__gender`) cites a "He" pronoun not present in the input. A post-hoc check verifying that quoted spans in the explanation appear verbatim in the input text would suppress this class of error without prompt changes. Complement with a "possible → positive" threshold relaxation analysis: item_recall_poss=0.615 on SynthPAI provides ~8 precision-estimable cases.

3. **Investigate multilingual/script-mixed OpenPII failures.** The 16–20% miss rate on OpenPII likely tracks items with non-Latin scripts, code-switched text, or name formats outside the judge's training distribution. Segment the OpenPII evaluation by script family to confirm.

4. **Treat FPs in fuzzy boundary categories as annotation review triggers.** Rather than counting FPs in `casual`, `troupe`, `weight` as judge errors, flag them as candidates for ground-truth review. The judge may be surfacing genuine privacy disclosures that initial annotation missed.

5. **Rate-limit the v5-vision run at ≤100 items/batch for OpenPII.** Run F's 48% error rate (48/100) was entirely rate-limit failures. With ≤100 items/batch and a 2–3 s inter-request delay, the same run would complete cleanly without the data loss.

Appendix A: Difficulty-Stratified Metrics (All Runs)

OpenPII

Difficulty	Evaluator	n	GT+	conf	TP	FP	recall	precision
explicit	v5 (Run A)	67	67	56	56	0	0.836	1.000
none	v5 (Run A)	133	0	30	0	30	N/A	0.000
explicit	v5-vision (Run F)	20	20	16	16	0	0.800	1.000
none	v5-vision (Run F)	32	0	4	0	4	N/A	0.000

HR-VISPR

Difficulty	Evaluator	n	GT+	conf	TP	FP	recall	precision
explicit	v5 (Run A)	210	210	37	37	0	0.176	1.000
none	v5 (Run A)	290	0	16	0	16	N/A	0.000
explicit	v5-vision (Run E)	71	71	53	53	0	0.746	1.000
none	v5-vision (Run E)	99	0	15	0	15	N/A	0.000

SynthPAI

Difficulty	Evaluator	n	GT+	conf	TP	FP	recall_lb	rec_poss	precision
explicit (location)	v5 (Run A)	6	6	0	0	0	0.000	0.667	N/A
implicit (identity)	v5 (Run A)	7	7	0	0	0	0.000	0.000	N/A
none	v5 (Run A)	237	0	0	0	0	N/A	N/A	N/A
explicit (location)	v5-vision (Run 3, pre-rev)	6	6	0	0	0	0.000	0.833	N/A
implicit (identity)	v5-vision (Run 3, pre-rev)	7	7	1	1	0	0.143	0.571	1.000
none	v5-vision (Run 3, pre-rev)	237	0	1	0	1	N/A	N/A	0.000

explicit (location)	v5-vision (Run 4, post-rev)	6	6	1	1	0	**0.167**	**1.000**	**1.000**
implicit (identity)	v5-vision (Run 4, post-rev)	7	7	0	0	0	0.000	0.286	N/A
none	v5-vision (Run 4, post-rev)	231	0	1	0	1	N/A	N/A	0.000

Appendix B: Attribute-Level Breakdown (HR-VISPR, v5-vision, Run E)

Attribute	n_ok	GT+	GT−	conf	TP	FP	recall	precision	coverage
age	8	8	0	2	2	0	0.250	1.000	0.250
casual	10	4	6	10	4	6	1.000	0.400	1.000
color	9	8	1	8	8	0	1.000	1.000	0.889
disability	11	0	11	1	0	1	N/A	0.000	0.091
ethnic_clothing	9	0	9	0	0	0	N/A	N/A	0.000
face	9	8	1	9	8	1	1.000	0.889	1.000
formal	11	3	8	3	2	1	0.667	0.667	0.273
gender	10	10	0	5	5	0	0.500	1.000	0.500
haircolor	11	10	1	10	10	0	1.000	1.000	0.909
height	6	3	3	0	0	0	0.000	N/A	0.000
medical	10	0	10	0	0	0	N/A	N/A	0.000
nudity	9	0	9	0	0	0	N/A	N/A	0.000
race	11	10	1	8	8	0	0.800	1.000	0.727
religion	10	2	8	1	1	0	0.500	1.000	0.100
sports	11	1	10	1	1	0	1.000	1.000	0.091
troupe	9	0	9	3	0	3	N/A	0.000	0.333
uniforms	8	0	8	1	0	1	N/A	0.000	0.125
weight	8	4	4	6	4	2	1.000	0.667	0.750

Appendix C: OpenPII Attribute Breakdown

v5 / gemini-2.5-flash-lite (Run A, n=200)

Attribute	n_ok	GT+	conf	TP	recall	precision	coverage
age	50	6	13	4	0.667	0.308	0.260
gender	50	3	15	2	0.667	0.133	0.300
identity	50	40	39	36	0.900	0.923	0.780

location	50	18	19	14	0.778	0.737	0.380

v5-vision / gemini-2.0-flash-001 (Run F, n_ok=52)

Attribute	n_ok	GT+	conf	TP	recall	precision	coverage
age	13	2	2	2	1.000	1.000	0.154
gender	11	1	2	0	0.000	0.000	0.182
identity	14	11	11	9	0.818	0.818	0.786
location	14	6	5	5	0.833	1.000	0.357